

Hsing-Hao Lee Curriculum Vitae

Contact Information

Department of Psychology, New York University

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Education

2021– 2026 Ph.D., Psychology (Cognition and Perception, Quantitative Concentration)

2021– 2025 Master of Philosophy, Psychology

Department of Psychology, New York University, USA

Mentor: Dr. Marisa Carrasco

2017– 2019 Master of Science, Psychology (GPA: 3.91/4.0)

Department of Psychology, National Taiwan University, Taiwan

Mentor: Dr. Su-Ling Yeh

2013– 2017 Bachelor of Science, Psychology (GPA: 3.94/4.0)

Department of Psychology, National Cheng Kung University, Taiwan

Mentor: Dr. Shulan Hsieh

Research Interests

1. **Attention:** How does visual attention modulate perception?
2. **Consciousness:** What are the differences between conscious and unconscious mental processes?
3. **Cognitive control:** How do people control and inhibit irrelevant information?

Publications and Manuscripts

1. Lee, H. H., & Carrasco, M. (2026). Covert spatial attention is uniform across cardinal meridians despite differential adaptation. *Journal of Vision*, 26, 15.
2. Yeh, S. L., Li, S. H., Lee, H. H., Kao, Y. W., Goh, J. O. S., Jingling, L., & Tsai, A. C. H. (*in press*). Cortical representation of apparent motion sequences in younger and older Adults: An fMRI study. *Scientific Reports*.
3. Chen, Q., Lee, H. H., Hanning, N.M., & Carrasco, M. (*preprint*). Transcranial magnetic stimulation to frontal cortex, unlike occipital stimulation, does not disrupt exogenous attention. *bioRxiv*.
4. Lee, H. H., & Carrasco, M. (2025). Visual adaptation stronger at horizontal than vertical meridian: Linking performance with V1 cortical surface area. *Proceedings of the National Academy of Sciences of the United States of America*, 122, (29) e2507810122.
5. Lee, H. H. (2025). Intrinsically photosensitive retinal ganglion cells and visual processing: ipRGCs beyond non-image-forming functions. *Frontiers in Neuroscience*, 19:1635101.

6. **Lee, H. H.**, Fernández, A., & Carrasco, M. (2024). Adaptation and exogenous attention interact in the early visual cortex: A TMS study. *iScience*, 27(11), 111155.
7. **Lee, H. H.**, Groves, K., Ripollés, P., & Carrasco, M. (2024). Audiovisual integration in the McGurk effect is impervious to music training. *Scientific Reports*, 14, 3262.
8. **Lee, H. H.**, Liu, G. K., Chen, Y. C., & Yeh, S. L. (2024). Exploring quantitative measures in metacognition of emotion. *Scientific Reports*, 14, 1990.
9. Cheng, T., Chiu, L., Huang, L., Lin, Y. T., **Lee, H. H.**, Chen, Y. C., & Yeh, S. L. (2023). Predictive processing in the “Second Brain”: From gut complex to meta-awareness. In *Expected Experiences* (pp. 170-194). Routledge.
10. **Lee, H. H.**, Chien, S. E., Lin, V., & Yeh, S. L. (2022). Seeing food fast and slow: Arousing pictures and words have reverse priorities in assessing awareness. *Cognition*, 225, 105144.
11. Chen, Y. T., **Lee, H. H.**, Shih, C. Y., Chen, Z. L., Beh, W. K., Yeh, S. L., & Wu, A. Y. (2022). An effective entropy-assisted mind-wandering detection system with EEG signals of MM-SART Database. *IEEE Journal of Biomedical and Health Informatics*, 26(8), 3649-3660.
12. **Lee, H. H.**, Chen, Z. L., Yeh, S. L., Hsiao, J. H., & Wu, A. Y. (2021). When eyes wander around: Mind-wandering as revealed by eye movement analysis with hidden Markov models. *Sensors*, 21, 7569.
13. **Lee, H. H.**, Tu, Y. C., & Yeh, S. L. (2021) In search of blue-light effects on cognitive control. *Scientific Reports*, 11, 15505.
14. **Lee, H. H.**, & Yeh, S. L. (2021). Blue-light effects on saccadic eye movements and attentional disengagement. *Attention, Perception, & Psychophysics*. 83, 1713-1728.
15. Chien, S. E., Chu, L., **Lee, H. H.**, Yang, C. C., Lin, F. H., Yang, P. L., Wang, T. M., & Yeh, S. L. (2019). Age difference in perceived ease of use, curiosity, and implicit negative attitude toward robots. *ACM Transactions on Human-Robot Interaction (THRI)*, 8(2):9.
16. **Lee, H. H.**, & Hsieh, S. (2017). Resting-state fMRI associated with stop-signal task performance in healthy middle-aged and elderly people. *Frontiers in Psychology*, 8:766.

Manuscripts in preparation

1. **Lee, H. H.** & Carrasco, M. (*in preparation*). The sequential effect of endogenous attention around polar angle.
2. **Lee, H. H.** & Carrasco, M. (*in preparation*). Does early visual cortex play a causal role in temporal attention? An TMS study.
3. **Lee, H. H.**, Kao, Y. W., Goh, J. O. S., & Yeh, S. L. (*in preparation*). Age-related differences in facial expression predictions: An fMRI study.

Selected Conference Presentations

1. **Lee, H. H.** & Carrasco, M. (2026). *Endogenous and exogenous attention remain uniform across cardinal meridians despite differential adaptation*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
2. Carrasco, M., Hanning, N. M., Fernández, A., Chen, Q., & **Lee, H. H.** (2026). *Cortical double dissociation for endogenous and exogenous covert spatial attention*. Talk presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
3. Chen, Q., **Lee, H. H.**, Fernández, A., Hanning, N. M., & Carrasco, M. (2025). *Neurostimulation reveals prefrontal cortex plays a central role in endogenous but not exogenous attention*. Poster presented at the Annual Meeting of the Society for Neuroscience 2025, San Diego, CA.
4. Carrasco, M., Chen, Q., **Lee, H. H.**, Fernández, A., & Hanning, N. M. (2025). *TMS to the prefrontal cortex disrupts endogenous, but not exogenous, attention*. Talk presented at the Annual Meeting of the European Conference on Visual Perception, Mainz, Germany.
5. **Lee, H. H.**, & Carrasco, M. (2025). *Endogenous attention enhances contrast sensitivity similarly around cardinal meridians despite differential adaptation effects*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
6. Chen, Q., **Lee, H. H.**, Hoxha, K., Fernández, A., Hanning, N. M., & Carrasco, M. (2025). *Does human right frontal eye field (rFEF+) play a critical role in exogenous attention? A Transcranial Magnetic Stimulation (TMS) study*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
7. **Lee, H. H.** & Carrasco, M. (2024). *Visual adaptation is more pronounced at the horizontal than vertical meridian*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
8. **Lee, H. H.**, Fernández, A., & Carrasco, M. (2023). *Adaptation modulates the effect of covert exogenous attention in early visual cortex – A TMS study*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
9. **Lee, H. H.**, Liu, G. K., & Yeh, S. L. (2021) *I know I'm happy, and I'm right: Metacognition of emotion*. Poster presented at the Annual Meeting of European Conference on Visual Perception, Virtual Meeting.
10. Cheng, T., Chiu, L., Huang, L. T., Lin, Y. T., Chen, Y. C., **Lee, H. H.**, & Yeh, S. L. (2021) Metacognitive confidence in perception, interoception, and memory. Poster presented at the Conference of Metacognition: New developments and challenges, Virtual Meeting.
11. **Lee, H. H.**, Chien, S. E., Lin, V., & Yeh, S. L. (2021). *Which comes first? Examining breaking continuous flash suppression time of high-calorie and low-calorie foods*. Poster presented at the Annual Meeting of the Vision Sciences Society, Virtual meeting.

12. Chen, Z. L., Lee, H. H., Chen, Y. T., Wu, A. Y., & Yeh, S. L. (2021). *The mind's eye: Mind-wandering revealed by eye movement hidden Markov model*. Poster presented at the Annual Meeting of the Vision Sciences Society, Virtual meeting.
13. Kao, Y. W., Lee, H. H., Goh, J. O. S., & Yeh, S. L. (2020). *Age-related differences in the statistical regularity of emotional faces*. Poster presented at the Annual Meeting of the Cognitive Neuroscience Society, Virtual meeting.
14. Lee, H. H., Shih, C. Y., Chen, Z. L., Wu, A. Y., & Yeh, S. L. (2019). *Are you concentrating or mind-wandering? Electroencephalogram signals reveal differences in attentional states*. Poster presented at the Annual Meeting of the Society for Neuroscience, Chicago, IL.
15. Lee, H. H. & Yeh, S. L. (2019). *Blue-light effects on the gap effect*. Poster presented at the Annual Meeting of the Vision Sciences Society, Tampa, FL.
16. Lee, H. H., Chen, Z. L., Chen, Y. T., Shih, C. Y., Beh, W. K., Wu, A. Y., & Yeh, S. L. (2019). *The heart never lies: Heart rate variability as an index of sustained attention*. Poster presented at the National Taiwan University-Kyoto University International Symposium for Cognitive Neuroscience, Taipei, Taiwan.
17. Lee, H. H. & Hsieh S. (2017). *The aging effect on time perception: An ERP study*. Poster presented at the 13th Asia Pacific Conference on Vision, Tainan, Taiwan.
18. Lee, H. H. & Hsieh S. (2016). *The relation between resting-state fMRI and stop-signal task performance in healthy middle-aged and elderly people*. Poster presented at the 55th Annual Taiwan Psychology Association Meeting, Tainan, Taiwan.

Research Experience

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| 2021 – | <p>Doctorate Student in Carrasco Lab</p> <p>Department of Psychology, New York University</p> <p>Principal Investigator: Dr. Marisa Carrasco</p> <p>Duties: Designed, programmed, collected and analyzed psychophysics, TMS, and fMRI experiments</p> |
| 2017 – 2021 | <p>Master's Student and Research Assistant in EPA (Explorer of Perception and Attention) Lab</p> <p>Department of Psychology, National Taiwan University</p> <p>Principal Investigator: Dr. Su-Ling Yeh</p> <p>Duties: Designed, programmed, collected and analyzed eye-tracking, fMRI, EEG, human-robot interaction and behavioral experiments</p> |
| 2016 – 2017 | <p>Research Assistant in CASE (Control, Aging, Sleep, and Emotion) Lab</p> <p>Department of Psychology, National Cheng Kung University</p> <p>Principal Investigator: Dr. Shulan Hsieh</p> <p>Duties: Collected and analyzed EEG, fMRI, neuropsychological tests and behavioral data from older adults</p> |

Awards and Honors

2025 – 2026	Taiwanese Overseas Pioneers Grant Issued by National Science and Technology Council (NSTC), Taiwan <i>PhD Candidate in Social Sciences</i>
2025, 2026	Ted Coons Graduate Student Travel Award Issued by Department of Psychology, New York University, USA <i>Vision Sciences Society, Florida, USA</i>
2023	Graduate School of Arts and Science Dean's Conference Fund Issued by New York University, USA <i>Vision Sciences Society, Florida, USA</i>
2021 – 2025	Taiwan's Government Fellowship for Studying Ph.D. Abroad Issued by Ministry of Education, Taiwan <i>Area: Cognitive Neuroscience</i>
2019	College of Science Student Travel Award Issued by National Taiwan University <i>Summer School Program at University of California, Berkeley, USA</i>
2019	Scholarship for Students Attending International Academic Conferences Issued by Ministry of Science and Technology, Taiwan <i>Vision Sciences Society, Florida, USA</i>
2017	Scholarship for Students Attending International Academic Conferences Issued by National Cheng Kung University <i>Asia-Pacific Conference on Vision, Tainan, Taiwan</i>
2016	College Student Research Scholarship Issued by Ministry of Science and Technology, Taiwan
2013 – 2016	Academic Excellence Awards (awarded every school year) Issued by Department of Psychology, National Cheng Kung University <i>Top 5% GPA among the school year in Psychology Department</i>

Teaching Experience

2026	Spring	TA for Undergraduate <i>Perception</i> , New York University
2025	Fall	TA for Undergraduate <i>Cognition</i> , New York University
2025	Summer	Instructor for Undergraduate <i>Cognition</i> , New York University
2025	Spring	TA for Undergraduate <i>Perception</i> , New York University
2024	Fall	TA for Undergraduate <i>Perception</i> , New York University
2024	Summer	Instructor for Undergraduate <i>Perception</i> , New York University
2024	Spring	TA for Undergraduate <i>Perception</i> , New York University
2023	Fall	TA for Graduate <i>Perception</i> , New York University

2019	Spring	TA for <i>Seminar in Graduate Institute of Psychology</i> , National Taiwan University
2018	Fall	TA for <i>Introduction to Psychology</i> , National Taiwan University
2017 – 2018		TA for <i>Statistics in Psychology and Education</i> , National Taiwan University
2017	Spring	TA for <i>Cognitive Psychology</i> , National Cheng Kung University

Graduate Course work

2024	Fall	Area Seminar: Perception & Attention
2023	Fall	Psychophysics
2023	Spring	Computational Cognitive Modeling
2023	Spring	Learning and Memory
2022	Fall	Mathematical Tools for Simulation and Data Analysis
2022	Spring	Functional Magnetic Resonance Imaging (fMRI) Lab
2022	Spring	Behavioral and Cognitive Neuroscience
2021	Fall	Perception
2021	Fall	Mathematical Tools for Neural and Cognitive Science

Ad-hoc Reviewer

Acta Psychologica, Attention, Perception & Psychophysics, Brain Sciences, Cognition, European Journal of Neuroscience, Frontiers in Human Neuroscience, Frontiers in Psychology, IEICE Transactions on Information and Systems, Journal of Eye Movement Research, Neuroscience of Consciousness, PloS One, Scientific Reports

Professional Affiliations

Asian-Pacific Conference on Vision (APCV), Cognitive Neuroscience Society (CNS), European Conference on Visual Perception (ECVP), Taiwanese Psychological Association (TPA), Society for Neuroscience (SfN), Vision Sciences Society (VSS)

Technical Expertise

R, MATLAB, Python, JASP, EEG (Neuroscan), fMRI (certified operator for Siemens Allegra 3T), Eye-tracker (EyeLink, Tobii), TMS (Brainsight, Magstim, MagVenture), Four-mirror stereoscope